



Module 01: PRAHARI

Lectures Stakeholder briefing · Amit Dua · September 2026

1.1 Who this document is for

This is the teaching note for PRAHARI. Readers are operators, architects, and stakeholders. The voice is a lecturer explaining a machine to a bright ten-year-old, then naming the real parts so an adult can inspect them.

Every number is one of three labels. MEASURED is a fact from this laptop or the Sentinel sandbox. DESIGN TARGET is statewide arithmetic we have not run at 80,000 cameras. CONJECTURED is unused in this note.

The working tree is 02_Code/prahari/. The product GitHub is amitduabits/PRAHARI. Collaborator: Amit Dua, Yushu Excellence Technologies Pvt. Ltd.

1.2 The problem, in one picture

Imagine 26 schools in one city. Each school has cameras. Each school bought a different cupboard to store the tapes. Some cupboards keep tapes for 7 days. Some keep them for 15. The head office has a notebook of stolen bicycles and missing children. Nobody in the camera room opens that notebook when a bicycle rolls past.

That is Gujarat's CCTV estate. Twenty-six departments. Analog and IP cameras. Sites from border districts to Valsad, Dahod, Somnath, Jamnagar, and Dwarka. Watchlists already exist: VAHAN, SARTHI, eGujCop, AFIS, NAFIS. The cameras do not match against those lists. There is no statewide census of which public-domain cameras exist, whether they are alive, or where coverage stops.

PRAHARI is a product of Yushu Excellence Technologies Pvt. Ltd. The collaborator is Amit Dua.

Key idea

The State does not need a new cupboard for every tape. It needs one notebook that watches the cameras it already paid for.

The design goal is a secure, scalable, interoperable, technically feasible, and cost-effective approach that uses existing infrastructure to the maximum practical extent. This PoC cannot replace 26 VMS contracts. A central recorder that ingests 80,000 full-HD streams (DESIGN TARGET) for 15 days needs statewide

dark fibre, petabytes of hot storage, and a GPU farm this tree does not have. PRAHARI therefore adds an intelligence plane on top of the systems that already record video.

1.3 The operating problem

Four hard facts, not four features to demo.

Heterogeneous infrastructure: vendors, VMS products, AMC contracts, analog and IP, H.264 and H.265, RTSP and HLS, 7-day and 15-day retention. A design that assumes one brand dies on day one.

Geographical dispersion of about 1,000 km. A border outpost and a Paldi junction do not share a LAN. UDP RTSP dies across NAT. The ingest contract therefore forces TCP.

Unified analytics and event handling: the same watchlist must fire on a Home camera and on a Transport camera. The event JSON must not change when the bus later becomes Kafka.

Onboarding without a redesign: a new department is a CSV plus a protocol adapter, not a rewrite of `analyse()`.

1.4 Architecture options, and the choice

Think of the options as four ways to organise a school.

Definition 1.4.1 (Model 1: registry and GIS). A roll call. Who owns the camera, where it sits, whether it is alive, how many days it keeps tape. No live picture is required.

Definition 1.4.2 (Model 2: unified viewing). One window that can open many classrooms. The school keeps its own cupboard. The window speaks RTSP, ONVIF, or a vendor API. ANPR may run on a sampled frame without storing every frame.

Definition 1.4.3 (Model 3: VMS federation). A post office between cupboards. Each vendor gets an adapter. Events travel on a bus. The original cupboard stays.

Definition 1.4.4 (Model 4: central VMS). One giant cupboard for the whole city. Ingest, store, playback, full AI, hot/warm/cold tape. This is a rip-and-replace.

Options we had		
Option	We take it?	Why
Model 1 only	No	A roll call does not catch a stolen car.
Model 2 only	No	Viewing without a watchlist fails the test case.
Model 3 full SDK zoo	No	Seven days cannot wrap every vendor SDK.
Model 4 statewide	No	We cannot host 80,000 recordings (DESIGN TARGET).
Hybrid 1+2+thin 3	Yes	Census now, viewing now, events now, recording later.

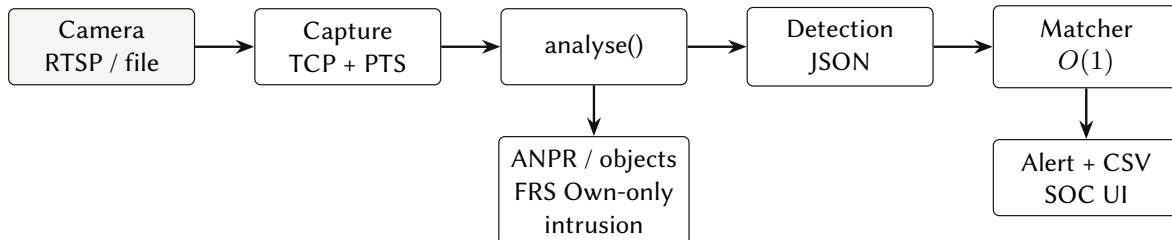
Definition 1.4.5 (PRAHARI hybrid). Model 1 supplies camera metadata, GIS, health, and gap reports. Model 2 opens reachable RTSP, HLS, WHEP, or ONVIF streams and runs analytics on sampled frames. Thin Model 3 is a detection-event JSON bus, an in-memory watchlist matcher, and `GET /api/track/{plate}`. Model 4 central recording is Phase-2 for selected cameras only.



A new vendor is one adapter. The detection JSON does not change if the bus later becomes Kafka. Kafka, Kubernetes, and Ceph are named as later swap points. They are not in this PoC.

1.5 How a picture becomes a red card

A ten-year-old can follow this path. An adult can grep it.

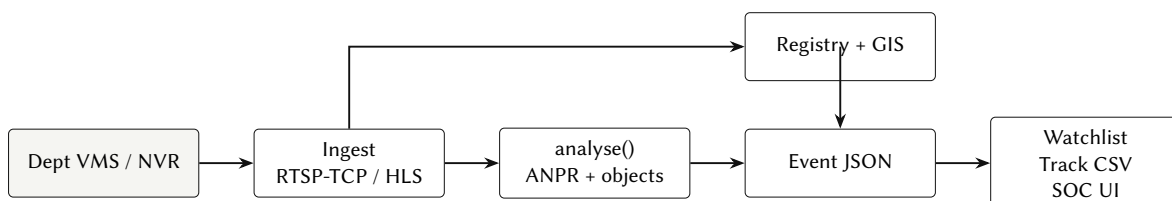


1. A camera produces a live stream, or a file clip for the own-feed demo.
2. Capture opens the stream over RTSP-TCP (or HLS, or a file). Time comes from PTS, not the wall clock.
3. analyse() looks at about one frame per second.
4. Each hit becomes a detection JSON row in SQLite.
5. The matcher checks a notebook (the representative watchlist) in $O(1)$.
6. A stolen plate raises a CRITICAL alert. The SOC tab turns red. A CSV records timestamps.

The stolen-car story

The notebook has GJ01AB1234 marked STOLEN (WL-001). Six seed cameras already saw that plate from Valsad to Gandhinagar. On 03 September 2026 an operator confirm on Paldi Circle (cam04) appended a seventh live point. The six seed ids stayed. The CSV is the operator artefact.

1.6 Architecture of the PoC



The PoC runs FastAPI and SQLite on one laptop. The UI is Leaflet plus vanilla JS on port 8080. Capture uses OpenCV with FFmpeg. Seven tabs: Operations, Cameras, Vehicle Track, Alerts, Watchlist, Onboard, Analytics & Gaps. There is no eighth tab.

PoC versus later swap		
Piece	PoC	Later swap
API	FastAPI + Uvicorn	same HTTP contract
UI	Leaflet + vanilla JS	React if a wall needs it
Store	SQLite	PostgreSQL + PostGIS
Capture	OpenCV + FFmpeg, RTSP TCP	DeepStream on regional GPUs
ANPR	morphology + Tesseract	YOLO plate detector + PaddleOCR
Objects	CPU blob / OpenCV DNN	YOLO11 at 1 fps on regional GPU
FRS	enrolled gallery, Own cameras	dedicated cameras, human confirm
Alerts	in-process WebSocket	Redis / Kafka
Auth	HTTP Basic + signed cookie	department SSO

1.7 Registry and GIS (Model 1)

A roll call needs a row per camera: id, department, ownership, consent, lat, lon, protocol, retention, health. Three doors put a row in: a CSV, a form, a REST call. A fourth door pulls the Sentinel catalogue.

Health is a colour on the Gujarat map. Three failed probes turn a pin red. Dahod GSRTC is seeded offline on purpose so a gap report has something true to say. Food and Civil Supplies cameras with 7-day retention appear in the short-retention list.

Private-permitted mall cameras require `consent=true`. A Home department viewer never sees that mall. That is the law of the notebook, not a CSS trick.

The live Sentinel catalogue has no coordinates. Those rows therefore do not sit on the map (the map already skips `lat = 0, lon = 0`). Operators open `cam04` from the Cameras table. The UI states that fact in words.

1.8 Live Sentinel contract

The sandbox publishes each camera as a live RTP/RTSP stream. One second of video takes one second to arrive. There is no rewind and no file download.

On 03 September 2026 the live portal is `https://cctv.corp8.cloud/`. Catalogue GET `/cameras.json`, after password login, returns a JSON array of `{id, name}`. Thirty cameras were listed. `/api/ingest` returns HTTP 404 on this host. PRAHARI therefore reads `/cameras.json`, never `/api/ingest`.

HLS playlists live at `https://cctv.corp8.cloud/<id>/index.m3u8`. The origin wants the access cookie and a browser User-Agent. RTSP is not on the TLS hostname. The live resource page publishes `rtsp://103.250.160.189:8554/stream/<id>`.

Warning

`/stream/<id>` answers range requests for a player. `curl` or `wget` of that path yields a partial file that looks complete. A team that ships that file as the government-feed demo has not consumed the grid. PRAHARI consumes live. It does not publish to the gateway.

MEASURED 03 September 2026: catalogue sync wrote 30 rows `cam01 ... cam30`. RTSP-TCP produced a live frame from `cam04` Paldi Circle, `1920×1080`, PTS 1080 ms. UDP SETUP failed with 461. TCP succeeded.



The manifest has no live flag and no lat/lon.

1.9 Live capture and the integrator laws

Why TCP, not UDP? UDP dies across NAT and most campus firewalls. Partial UDP looks like a broken model. The capture client sets `OPENCV_FFMPEG_CAPTURE_OPTIONS=rtsp_transport;tcp` before it imports OpenCV. If port 8554 is blocked, the client falls back to HLS.

Why PTS, not the wall clock? After connect, the gateway may replay a group of pictures faster than real time. A tracker that stamps arrival time invents impossible speeds. Event time is `CAP_PROP_POS_MSEC`. Declared FPS is unused.

Why backoff 2 s to 30 s? Feeds restart. A tight reconnect loop is a denial of service against ourselves.

Why a scene cut? Each sandbox feed loops. At the loop the picture jumps. Object track ids and face tracks reset. A PTS rewind or a gap above 5 s is that jump.

Why a cap of four open captures? A laptop is not a regional GPU site. `MAX_OPEN_CAPTURES` defaults to 4. The fifth open raises. `SCALE_BENCH MEASURED` that fact on 04 September 2026.

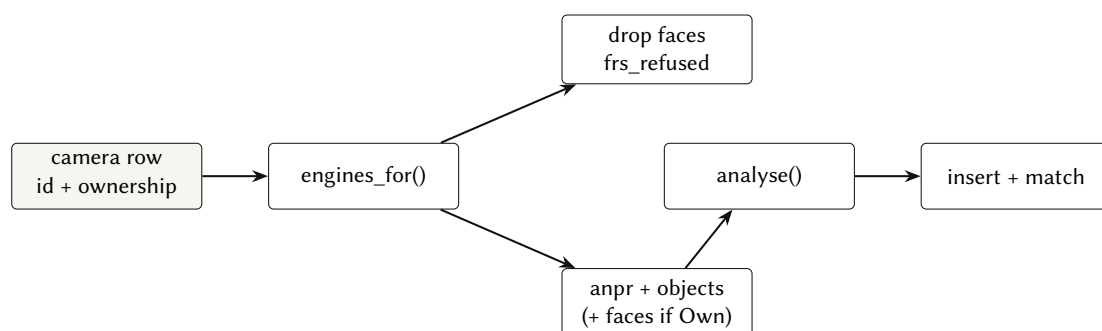
Implementation

`tests/test_integrator_laws.py` greps the capture client for the TCP flag, forbids `CAP_PROP_FPS` in services, and forbids publish helpers. `test_scene_cut.py` checks rewind and gap, and checks that object track ids reset.

1.10 analyse(): one door for every engine

A child should not have four different “look at this picture” buttons that disagree. Adults should not either.

Definition 1.10.1 (`analyse(frame, camera, pts_ms)`). The only worker entry. It returns a list of detection dicts. It does not insert them. The ingest router writes crops, inserts rows, and calls the matcher.



Default engines are anpr, objects. Own cameras may also list faces. Government catalogue ids matching cam plus digits never receive the faces engine. Ownership Gov never receives it either. The log line is `frs_refused`. FaceAnalyzer is not constructed on those cameras. Deleting person events after the fact is not the control. The engine is never started.

Godown camera CAM-FCS-001 additionally runs intrusion: a person box that overlaps a stored ROI by at least 30 percent of the person area.

Plate fields stay frozen so old callers keep working. Additive fields: entity type and id, face id, object class, bbox, track id, source. Crop honesty fields: original uri, optional enhanced uri, enhancement method, reconstructed flag.

Implementation

POST /api/ingest/frame stays ANPR-only. POST /api/ingest/analyse is the multi-engine door. Onboard has both buttons. Tests post a face-like still to cam04 through /analyse and expect zero person events.

1.11 ANPR: options and the choice

A number plate is a name tag glued to a car. We need the letters in a standard spelling so GJ 01 AB 1234 and GJ-01-AB-1234 are the same name.

Definition 1.11.1 (Indian plate normaliser). After stripping spaces and hyphens, the string must match $^{[A-Z]\{2\}[0-9]\{1,2\}[A-Z]\{1,3\}[0-9]\{3,4\}}\$$. Anything else is dropped.

ANPR options		
Option	Taken?	Constraint
Train a detector from scratch	No	Seven days, no labelled Gujarat set.
Require a GPU YOLO now	No	The laptop has GPU count MEASURED 0.
Tesseract + morphology	Yes (default)	CPU, swap behind recognize().
YOLO vehicle crop then Tesseract	Optional	ANPR_ENGINE=yolo, falls back.
Operator confirm	Yes	A readable plate cannot fail a live demo.

recognize(frame_bgr) returns {plate, plate_raw, confidence, crop}. Callers never import Tesseract. If the Tesseract binary is missing, the function returns an empty plate. It does not crash.

When ANPR_ENGINE=yolo, the backend asks YOLO for a vehicle box, crops it, optionally asks a plate model, then runs Tesseract on that crop. Any import or empty-box failure falls back to morphology plus Tesseract on the full frame. Default remains tesseract.

Operator confirm writes confidence 1.0 and source=operator_confirm. A confirm row is never labelled source=anpr. That honesty is a test, not a slogan.

MEASURED 03–04 September 2026: Tesseract was not on PATH. Confirm on cam04 wrote GJ01AB1234. Eight own-feed stills at PTS 1 s produced no plate. The log says so.

1.12 Object detection: options and the choice

A person is a blob that looks like a person. A car is a blob that looks like a car. Operators need person and vehicle detection, not only plates.



Object-engine options		
Option	Taken?	Constraint
YOLOv8n as the only path	No	Extra dependency, tests would skip.
OpenCV DNN with vendored weights	Later	Weights are large. Tests must not need a download.
Skin-tone blob fallback	Yes for tests	A drawn rectangle is a person if it occupies $\geq 8\%$ of the frame.
YOLO behind OBJECT_ENGINE	Optional	Person class 0 is included. Fallback is blob.
Skip objects entirely	No	Eval area 05 names them.

Classes kept: person, car, motorcycle, bus, truck, bicycle. Others drop. Each box gets a per-camera track_id. Default assignment is IoU. TRACK_ENGINE=bytetrack uses ByteTrack when ultralytics loads. A scene cut still calls objects.reset, which also resets ByteTrack state.

Object rows do not raise a watchlist card unless they are wrapped as intrusion.

GET /api/objects/report.csv is the operator artefact: timestamp, camera, class, confidence.

Weights live at app/models_data/yolov8n.pt and are gitignored. scripts/fetch_yolo_weights.py downloads them once. Pytest does not download them. Default pytest stays green without ultralytics.

1.13 Lawful FRS: options and the choice

A face is a name tag a person did not pin on. Matching faces of strangers on a public street is a rights problem, not a prize feature.

FRS options		
Option	Taken?	Constraint
No FRS at all	No	HLD text and eval 05 name recognition.
Wild FRS on Paldi Circle	No	Unknown members of the public. Unlawful in this PoC.
Live AFIS / NAFIS pipe	No	We have no ministry API. Claiming one is a false claim.
Enrolled gallery, Own cameras	Yes	Consented adults or synthetic fixtures. Human confirm.
FaceNet 512-d vectors	Optional	FACE_ENGINE=facenet, Own only, lazy import.

Warning

The faces engine is refused when ownership is not Own, and when camera_id matches cam plus digits. Tests post a face-like still to cam04 and assert zero person events. A spy on FaceAnalyzer.__init__ must stay at zero calls. The government-feed demo never shows FRS.

The gallery stores crops under data/faces/{gallery_id}/. Default match uses a colour histogram of a

64×64 crop. GET /api/faces/gallery returns ids and counts. It does not return embeddings. WL-004 is a person row. The matcher fires on face_id, not on an empty plate.

Synthetic fixtures are geometric ellipses with unique noise. Tests do not need a photograph of a real person. A human demo may add two consented adult photos. Never a minor. Never a crop from cctv.corp8.cloud.

1.14 Optional vision engines

FaceNet, YOLO, ByteTrack, next-camera predict, and a person-enroll form sit behind env doors in this tree. Defaults need no GPU.

Definition 1.14.1 (Engine env). Four getenv keys select backends. Defaults need no GPU. Optional values load only when extras exist.

Key	Default	Optional
ANPR_ENGINE	tesseract	yolo
OBJECT_ENGINE	opencv	yolo
FACE_ENGINE	histogram	facenet
TRACK_ENGINE	iou	bytetrack

ANALYTICS_ENGINES defaults to anpr,objects. Add faces only for Own cameras.

FaceNet path: MTCNN boxes plus InceptionResnetV1 512-d embeddings, cosine against .npy files next to gallery JPEGs. Torch is imported inside FaceAnalyzer.__init__, never at package import. If torch is missing, match() falls back to histogram and does not crash.

A naive ingest that constructed FaceAnalyzer on every camera is refused. engines_for drops faces before match() is called. That is the FRS law, not a filter after the fact.

OBJECT_ENGINE=yolo includes COCO class 0 (person) so object reports cover people as well as vehicles.

Branded reconstructor names are not in app/. Optional enhanced crops are labelled cubic_upscale or none. Matcher always sees original pixels.

Implementation

requirements.txt stays FastAPI, OpenCV, Tesseract, pytest. Vision extras live in requirements-vision.txt. A CPU-only workstation that skips that file still runs the demo. GPU count MEASURED on 04 September 2026 is 0.

1.15 Crop honesty

Every persisted detection writes the original JPEG uri twice (legacy field and crop_uri_original). An optional enhanced copy is a cubic upscale of a low-FFT crop, for a human to look at. Auto-match uses original pixels only.

If a face crop is marked is_ai_reconstructed=1, the matcher inserts status=pending_review and does not push a CRITICAL WebSocket card from that crop. A reconstructed face is a review item, not an automatic arrest card.



1.16 Intrusion on a godown

Food and Civil Supplies cameras watch a warehouse, not a highway. A person standing in a drawn rectangle is an intrusion.

The ROI lives in `cameras.extra_json`. Overlap of at least 30 percent of the person box with the ROI raises `entity_type=intrusion`, category `INTRUSION`, priority `CRITICAL`. Two hits within 120 s on the same camera increment a counter. There is no fourth neural net. Intrusion wraps object boxes.

1.17 Watchlist, alerts, track

The notebook is a representative watchlist. It is not a live ministry pipe. Categories include `STOLEN`, `BLACKLIST`, `OBSERVE`, and a person row `MISSING_ASSOCIATE`.

The matcher keeps plates in a dictionary and gallery ids in another. Lookup is $O(1)$. Reload is on demand (and on boot). Duplicate suppression: the same entity on the same camera within 120 s increments counter. Priority `CRITICAL` is stolen, wanted, or intrusion. `HIGH` is blacklist or the person associate. `LOW` is log.

`WL-001` cannot be deleted. That is the seeded demonstration plate.

The Watchlist tab has an Enroll Missing/Wanted Person form. It posts `/api/faces/enroll`. A `home.viewer` still receives 403. Alerts that sit in `pending_review` show that word. Ack still works.

Designated vehicle GJ01AB1234

`GET /api/track/GJ01AB1234` returns the six seed points in this order: CAM-VAL-001 Valsad 06:12, CAM-SUR-001 Surat 07:48, CAM-AHD-004 Narol–Naroda 09:05, CAM-AHD-001 SG Highway 09:22, CAM-GNR-003 Koba 09:51, CAM-GNR-001 Gandhinagar Sector 21 10:04. Live confirm on `cam04` appended. CSV columns include plate, camera, timestamp, location.

Cross-camera tracking in this PoC is plate sightings joined to camera GIS. It is not a Re-ID embedding across Paldi Circle. A teleport flag fires if implied speed exceeds 180 km/h. That flag is information, not a hard fail. ByteTrack `track_id` is per camera. It does not replace `GET /api/track/{plate}`.

Alerts push on WebSocket `/ws/alerts`. The UI also polls every 5 s so a blocked socket does not hide a `CRITICAL` card. Person cards show a name or gallery id, not a blank plate.

1.18 Next-camera predict and keyword query

`GET /api/predict/{plate}` ranks likely next cameras. First it counts historical transitions out of the last camera that saw the plate. If that map is empty, it ranks GIS neighbours by haversine distance. The method field is `frequency+distance`. It is not Kalman. It is not Re-ID. The Track tab has a Next cameras button.

`POST /api/query` extracts a plate, a camera id, or an object class with regular expressions. The response includes `engine: keyword_rules`. It is not a language model. Slides must not call it NLP.

1.19 Seven tabs, and how a judge walks them



Operations: Gujarat map, health-coloured pins, video wall of at most four tiles. HLS uses vendored `hls.min.js`. Stream URLs carry a 60-second HMAC token. Raw `rtsp://` never reaches the browser.

Cameras: table with Open tile. Live catalogue rows have no coordinates. The copy says so.

Vehicle Track: reconstruct GJ01AB1234, download CSV, optional Next cameras.

Alerts: CRITICAL queue, pending_review word, Ack.

Watchlist: representative rows, person enroll form.

Onboard: add camera, CSV import, ANPR this still, Analyse this still, operator confirm plate, confirm face.

Analytics & Gaps: object CSV, entity counts, short retention, Dahod offline.

Footer: DESIGN TARGET at 80,000 cameras we sample 1 fps at the region. MEASURED cap is four open tiles.

1.20 Security and privacy

Roles: superadmin, soc_operator (judge), dept_viewer, auditor. A Home viewer sees Home cameras only. An auditor cannot POST. Stream tiles use a 60-second HMAC token. Responses to the browser omit `rtsp://`. The Sentinel cookie stays on the server.

Path jail keeps crops inside the media root. HLS origin is pinned to the configured Sentinel host. Passwords compare with `hmac.compare_digest`.

Audit rows cover onboard, watchlist edit, ack, report download, plate confirm, face enroll, and face confirm. Stream URLs are not stored in audit detail.

FRS is not AFIS and not NAFIS. Gallery JSON has no embeddings. Government CCTV of unknown people never runs FRS.

1.21 Scale and cost

A ten-year-old version: do not ship every school video to one room. Ship a tiny still once a second, and only from cameras that matter. The wall shows 16 to 64 live tiles. The rest is metadata.

Numbers below mix MEASURED laptop facts with DESIGN TARGET statewide arithmetic.

Bandwidth arithmetic (DESIGN TARGET)

Take 45,000 public-domain cameras, 1 fps, 80 KB JPEG crop (DESIGN TARGET). Bytes per second $= 4.5 \times 10^4 \times 8 \times 10^4 = 3.6 \times 10^9$, which is 3.6 GB/s if every crop flies to one room. Five regions divide that by five: about 720 MB/s each. The SOC wall still pulls only 16 to 64 live tiles. Metadata events are hundreds of bytes, not an 80 KB crop.

Assume 45,000 public-domain cameras run analytics at 1 fps with an 80 KB JPEG crop (DESIGN TARGET). Naive centralisation is $45,000 \times 80 \text{ KB/s} \approx 3.6 \text{ GB/s}$. Five regional GPU sites (Ahmedabad, Surat, Rajkot, Vadodara, Bhuj) cut that to about 720 MB/s per site. Low-connectivity outposts send events only.



MEASURED on 04 September 2026 on this laptop: four open file captures succeed; the fifth is rejected. Mean JPEG from eight own-feed stills is 41 KB. /api/health p99 is 9.4 ms over 50 calls. GPU count is 0. Statewide 80 KB and regional GPUs remain DESIGN TARGET.

Order-of-magnitude yearly cost of the intelligence plane, not of replacing 26 VMS estates (DESIGN TARGET): GPU nodes 1.8–2.4 Cr, object storage 0.8–1.2 Cr, bandwidth 0.6–1.0 Cr, twelve engineers 1.5 Cr; total about 5–6 Cr.

Hot / warm / cold recording of every frame is Model 4 and Phase-2, selected cameras only.

1.22 Test cases

The suite on 04 September 2026 with default engines and no torch: 88 passed, 4 skipped. Skips are Tesseract binary, FaceNet extras, YOLO weights, ByteTrack. scripts/audit_gate.py prints PASS.

C-suite families		
Family	Files	Claim
Integrator	test_integrator_laws, test_scene_cut	TCP, no FPS timing, scene-cut reset
Registry	test_cameras, test_catalogue	CSV, consent, catalogue 400 without host
ANPR	test_plate_normaliser, negatives, honesty	Confirm is not ANPR. Junk drops.
Objects	test_objects	Person blob inserts. YOLO without weights does not crash.
FRS	test_faces, test_privacy_frs	Same gallery matches. cam04 emits zero faces.
Intrusion	test_intrusion	ROI overlap CRITICAL. Missing ROI is a no-op.
Matcher	test_matcher	Stolen, blacklist, WL-004 without a plate, 120 s dedupe
Track	test_track	Six seed cameras. Append does not drop seed-1.
Security	test_security, path jail, no RTSP leak	Viewer 403 on enroll. Auditor 403 on confirm-face.

Vision pack T-V		
ID	File	Claim
T-V01	test_engine_fallback	FaceNet without torch falls back to histogram
T-V02	test_engine_fallback	YOLO without ultralytics falls back to blob / Tesseract
T-V03	test_facenet_optional	Skip unless torch. Same gallery cosine beats other.
T-V04	test_privacy_frs	FaceAnalyzer not constructed on cam04
T-V05	test_yolo_optional	Skip without weights
T-V06	test_bytetrack_optional	Skip without ultralytics. Reset clears state.
T-V07	test_crops_honesty	Original uri set. Reconstructed is pending_review.
T-V08	test_predict	Predict returns a list. Never 500. Auth required.
T-V09	test_query_rules	engine=keyword_rules. Not nlp, not llm.
T-V10	test_enroll_ui	Index contains Enroll. Viewer 403.
T-V11	test_no_forbidden_model_names	Zero branded reconstructor names in app/

1.23 Experiments

scripts/run_experiments.py -suite smoke appends 05_Output/experiments/EXPERIMENT_LOG.md. Each row is MEASURED or DESIGN TARGET or SKIPPED with a reason.

E-A1 ANPR fixture SKIPPED without Tesseract. E-O1 person blob MEASURED. E-F1 histogram gallery hit MEASURED. E-I1 intrusion MEASURED. E-W1 track count ≥ 6 MEASURED. E-S1 fifth session rejected MEASURED. E-S6 GPU 0 MEASURED. E-G1 government live grab SKIPPED when SENTINEL_HOST is empty.

E-V1 histogram FRS still matches WL-004. E-V2 FaceNet SKIPPED without torch. E-V3 YOLO SKIPPED without ultralytics. E-V4 cam04 plus FACE_ENGINE=facenet still refuses faces, MEASURED. E-V5 predict returns a list, MEASURED.

1.24 Capture internals

StreamSession is the only object that holds an OpenCV capture. The TCP transport flag is set in the environment before OpenCV is imported. Open counts against MAX_OPEN_CAPTURES (default 4). The fifth open() raises. That is a laptop limit, not a statewide SLO.

read() returns (ok, frame, pts_ms). pts_ms comes from CAP_PROP_POS_MSEC. Wall clock is used only inside backoff_sleep. A decoder warning at join is logged. It does not abort the session. HLS origin is pinned to the configured Sentinel host. File protocol is for own_feed.mp4 on CAM-OWN-001.

1.25 Catalogue client

app/services/catalogue.py logs into the Sentinel host with a session cookie, then GET /cameras.json. Ids come from JSON id fields. The live host has no coordinates and no /api/ingest. A missing SENTINEL_HOST is valid: the PoC boots on sample cameras. Tests assert 400 when a sync is requested without a host.



1.26 Matcher worked example

Boot loads WL-001 plate GJ01AB1234, STOLEN, CRITICAL. A detection on CAM-VAL-001 at 06:12 inserts alert *A* with counter=1. A second detection on the same camera 10 s later increments *A* to 2. A detection 3 minutes later inserts alert *B*. A detection of XX99YY9999 inserts nothing.

A person detection with face_id=WL-004 and empty plate inserts a HIGH person alert. A reconstructed face with is_ai_reconstructed=1 inserts pending_review and does not notify CRITICAL.

1.27 RBAC

Four users ship in .env.example.

User	Role	Department	Write?
judge	soc_operator	all	yes
admin	superadmin	all	yes
home.viewer	dept_viewer	Home	no (403)
auditor	auditor	all	no (403)

home.viewer never sees CAM-MALL-001. Enroll and confirm-face are write routes. Tests cover 401 without auth and 403 for viewer/auditor.

1.28 Path jail and stream tokens

app/paths.py resolves crop and HLS files against allowed media roots. A `..` path is rejected. Stream tiles use `issue_stream_token` / `verify_stream_token`: HMAC over camera id, actor, expiry. TTL default 60 s. A sliced hex digest is forbidden; audit_gate S4 checks that. The browser never receives `rtsp://`. `hls.min.js` is vendored. `jsDelivr` is forbidden in `index.html`.

1.29 HTTP surface

Write routes require `assert_write`. GET `/api/health` is public.

Path	Fact
POST <code>/api/ingest/frame</code>	ANPR only
POST <code>/api/ingest/analyse</code>	multi-engine door
POST <code>/api/ingest/confirm</code>	source=operator_confirm
POST <code>/api/ingest/confirm-face</code>	Own cameras only
GET <code>/api/track/GJ01AB1234</code>	six seed points, append-safe
GET <code>/api/predict/{plate}</code>	frequency then GIS
POST <code>/api/query</code>	engine=keyword_rules
POST <code>/api/faces/enroll</code>	viewer 403
GET <code>/api/faces/gallery</code>	no embeddings
GET <code>/api/objects/report.csv</code>	operator artefact

1.30 Repository map



02_Code/prahari/app/services/analyse.py is the only worker. anpr.py, objects.py, faces.py, intrusion.py sit behind it. app/engines/ holds lazy FaceNet, YOLO, ByteTrack. app/routers/ is HTTP. app/static/ is seven tabs plus vendored HLS. tests/ is the C-suite plus T-V. scripts/audit_gate.py is the claims lock. scripts/run_experiments.py writes MEASURED rows. 04_Documents/ holds this note, the deck, and the HLD.

1.31 Failure modes the tests already cover

Missing Tesseract: empty plate, no crash. Missing torch: histogram FRS. Missing ultralytics: blob objects and Tesseract ANPR. Fifth live session: 429. Scene cut: track ids reset. Gov camera plus faces in the form: zero person events. Viewer enroll: 403. Tampered password: lookup fails. Path jail: . . rejected. Confirm: source is never anpr.

1.32 How to run the PoC

Clone amitduabits/PRAHARI. From 02_Code/prahari copy .env.example to .env. Set the SOC operator password before any hosted demo. Set the catalogue host, password, and RTSP IP after login. Run run.ps1 (Windows) or run.sh (Unix). Open <http://127.0.0.1:8080> and sign in as judge.

Pytest from that folder: `python -m pytest -q`. Gate: `python scripts/audit_gate.py`. Experiments: `python scripts/run_experiments.py --suite smoke`. Logs land in 05_Output/experiments/.

TeX: from 04_Documents/tex run `pdflatex slides.tex` twice and `pdflatex notes.tex` twice. Copy PDFs to 04_Documents/ and docs/.

Optional vision: install requirements-vision.txt on the demo laptop only. Then set `FACE_ENGINE=facenet` or `OBJECT_ENGINE=yolo`. Defaults need no GPU.

Three-minute walk: Operations map (seeded pins, Dahod offline), Cameras table Open tile on cam04, Vehicle Track reconstruct GJ01AB1234 and download CSV, Alerts CRITICAL and Ack, Onboard Analyse this still or confirm if OCR misses, Analytics & Gaps for short-retention Food and Civil Supplies cameras.

1.33 Capability map

Government-feed ingest: catalogue cameras, RTSP-TCP frame, confirm plus CSV on cam04. Architecture: this note, the Beamer deck, and the HLD; open adapters; central recording not faked. Working platform: FastAPI on :8080. Analytics: ANPR/confirm, objects CSV, godown intrusion, Own-only gallery FRS. Scale: 1 fps regional DESIGN TARGET in §1.21. Optional: hybrid track, objects, intrusion, lawful FRS, RBAC, gaps, WebSocket.

1.34 What this tree contains

Hybrid plane, catalogue sync, RTSP-TCP frame, confirm CSV, objects, intrusion, lawful FRS with government-camera refuse, optional FaceNet/YOLO/ByteTrack behind env doors, predict, keyword query, RBAC, experiment harness, this deck.

Tesseract on PATH is optional; operator confirm still writes a detection. Two consented adult photos are optional for a live face demo. FaceNet own-feed still is SKIPPED on a laptop that has no torch.



Government-camera FRS remains forbidden.

